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Automatic Colorization of Greyscale Images Using Deep Learning Fusion Technique and Accuracy Level Analysis

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***Abstract*— Recent technological advances have opened new ways to colorize grayscale images automatically, using different deep-learning techniques. Automatic colorization plays a crucial role in various applications such as coloring low-light or night vision images, historical photo restoration, for lights-out factories, medical imaging, and many more. Storing images in grayscale format significantly reduces storage requirements, making it an efficient solution for media storage and transmission, such as in CCTV footage, where grayscale can save up to two-thirds of storage space compared to color. Hence, grayscale image colorization has far more importance than anticipated. Colorization techniques were studied through an extensive literature review. Then, error calculations of colorized images were compared and analyzed. Progressively, different parameters which affect the colonization were changed and compared as hyperparameter tuning. Building upon this knowledge, an improved algorithm using a fusion-based approach was developed. The algorithm concatenates local and global features of the image for plausible colorization results. The model was validated by using different images and is quantitively as well as qualitatively compared. Seven types of error calculation methods are used in this paper for comparison of the original and the predicted images. Furthermore, due to the lack of quantitative visual representation for the accuracy analysis of colorized images, a new concept of error analysis using heat maps are proposed.**

I. Introduction

Greyscale images consist of shades of gray or either black or white, representing the intensity of the pixel. The process of colorization involves adding Red, Green, and Blue (RGB) color information to each pixel. Potential applications are for coloring historical images, night vision, military purposes, low light conditions, saving image storage space, saving energy in factories, and in the medical industry. To protect the diversity of cultures, it is essential to enhance cultural heritage in historical photographs [1]. Since color is an integral part of identifying objects, law enforcement authorities and others can use this information to trace back the image chain in CCTV footage [2]. Furthermore, images in computers are represented by binary numbers. The bits per pixel takes up more space in RGB, requiring 24 bits (8 bits for each RGB channels), while greyscale needs only 8 bits for brightness ranging from 0 (black) to 255 (white) [3]. Hence, to address bandwidth and storage constraints, video or images can be stored in greyscale, which can later be converted to color using reliable tools. For instance, a 1024x1024 colored pixel image takes 3MB, while greyscale requires only 1MB. This results in two-thirds storage savings. It is particularly beneficial for CCTV footage, where greyscale storage can extend the recording capacity from one to three weeks. Another application is that factories focus on energy efficiency by adopting lights-out robotic systems. An example is demonstrated by FANUC since 2001 [4]. This approach has enhanced productivity and reduced upkeep costs. Additionally, this can be applied to medical imaging, where existing radiological display techniques only show grayscale mammography images [5]. However, colorization relies on color spaces, which can be interchanged using standard formulas [6]. The RGB color space is closely related to the human visual system. It is also the most widely used color space in computer vision technologies [7]. The methods for colorization can be broken down into scribble-based, example-based, and learning-based [8]. Despite significant advances in image processing and deep learning techniques, predicting colors for grayscale images remains a challenging task [9].

A. Using Auto-Encoder Models

A novel colorization Deep Learning Convolutional Neural Network (DLCNN) has been proposed using the ImageNet dataset [10]. RGB images have been transformed into the Lab color space feeding the network with grayscale input L, which then predicts a and b channel values normalized to 313 bins [11], [12]. Performing convolution on a single channel grayscale image at each pixel position has produced the corresponding color value [13]. A model using Keras with TensorFlow has implemented a feed-forward CNN, by training for more than 20000 color images [14]. An automatic learning-based colorization algorithm based on the VGG-16 CNN has been proposed by [15]. Various other CNN models have addressed the colorization problem, including one trained on over a million images using an image classifier [8]. Another model has treated colorization as a classification task with class rebalancing [16]. Systems like [17], [18] leverage large-scale data and CNNs with distinct architectures and loss functions. [17] has used an un-rebalanced classification loss, employing hype columns on a VGG network, while [18] has used a regression loss with a two-stream architecture, integrating global and local features for automatic colorization. The model has been trained for 3 weeks on the Places dataset, comprising 2,448,872 training images. Image Colorization using CNNs and Inception-Resnet-v2 by [19] is another merged model that proposes an Autoencoder (AE) for image colorization. The features from the AE are fed through the Inception-ResNet-v2 network, which generates color information. The network has aimed to capture multi-scale features by using parallel convolutional operations with different kernel sizes.

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B. Generative Adversarial Networks (GANs)

An algorithm uses the U-Net network to learn image colour information and build a multi-layer convolution to form a discriminative network for adversarial training [20]. They have used GANs to train on the CIFAR-10 and Places365 dataset, and its results have been compared with those obtained using existing CNN. One of the results sets has shown the Sepia effect due to the L2 regularization that caused blurring. Another paper has defined a conditional GAN for image-to-image translational problems [21], commonly known as pix2pix. For the generator, they have used a 'U-Net' based structure. A PatchGAN classifier has been used for the discriminator part.

C. Error Analysis

In the context of error analysis, several studies have relied on human validation to assess the accuracy of colorization processes [18], [22]. In contrast to human validation methods, automated assessments tend to have discrepancies. The results from [12] and [20] appear desaturated, attributed to loss functions inducing conservative predictions, resulting from standard regression problems. The objective has been to minimize the Euclidean error between an estimate and the actual value [16]. Various techniques evaluate generated image values through Perceptual Realism, Semantic Interpretability, and Raw Accuracy. Whereas some have used quantitative evaluation metrics such as Peak Signal Noise Ratio (PSNR), Mean Square Error (MSE), and Average inference time to measure the accuracy between the original and predicted images [11].

Automatic colorization of grayscale images and accuracy analysis is a complex research area, made challenging by the difficulty in accurately identifying and applying appropriate colors to grayscale images. While numerous research studies have attempted to address this issue, no system has been developed to colorize grayscale images with 100% accuracy. Accuracy levels have been measured and reported by different parameters in various research studies. However, previous research has not provided adequate techniques to visually demonstrate the errors in the predicted images. This is a significant limitation, making it difficult to identify areas for improvement in the predicted colorized image. As a result, there is a need for a more common, comprehensive, and easily understandable approach to analyzing the accuracy levels of automated colorization systems. Therefore, the research aimed (1) to enhance existing automatic colorization methods and (2) to analyze a quantitative technique to visually demonstrate errors in the predicted images.

II. Methodology

A. Models Used for Colorization

Two methods, AE [23] and GAN models [24] were selected. Evaluation involved testing with three datasets, varying epochs, and quantitatively comparing error values in R, G, and B channels for choosing the best method for further analysis. PSNR, Average error magnitude per pixel, and normalized MSE were computed based on two categories: Test1, evaluate colorization by using images similar to the dataset and Test2 - evaluate colorization by using images available in the dataset. Initially, a small dataset was used to run the AE and GAN models aiming to save computation time and ensure code accuracy. As the code improved, the need for a larger and more relevant data set emerged. The initial aim of the research was to colorize any natural greyscale image due to its constant consistency in colors unlike man-made objects. However, the computational inefficiency with the given processing power narrowed the research to focus solely on colorizing natural sceneries in green and blue. This led to the creation of a training dataset of 346 images using [25], [26], [27], and [28].

B. Hyperparameter Tuning

This research compares parameters such as training batch sizes, iterations, optimizers, number of convolutional layers, filters, and convolution layer output sizes, with detailed results. Achieving a balanced number of convolutional layers is critical for avoiding overfitting and managing computational complexity. Hence, a trial-and-error approach was employed to determine the optimal use of convolutional layers. [25], [26], [27], and [28].

C. Improved Algorithm

CNNs allow the extraction of global or high-level and local or low-level features [18]. Global features capture overall context, abstract information, and large-scale patterns, while local features emphasize fine details, textures, basic elements like edges. The combination enables CNNs to effectively learn colorization mappings, producing realistic, visually pleasing images. Global features include contour representations, shape descriptors, texture, brightness levels influenced by the sunset capturing overall spatial relationships, such as the sun's position relative to mountains in a sunset image [29] and [30]. These features are automatically recognized by deep learning models through architectural design and training. Local features include distinct mountain peaks, clouds, trees, and fine details such as edges, contours, and internal textures of leaves, grass, or flowers. Hence, to improve basic AE algorithm, a new approach combines a convolutional autoencoder with a merge model. The encoder compresses the input image making it dimensionally smaller. The more the encoder encodes, the more features the model can extract [31]. That means, by increasing the depth of the encoder model with layers. But the limitation is, the further we reduce the dimension of the encoder model, it is hard for the decoder model to regenerate the required output. So here the encoder model has been split into 2, where one path extracts more features taken from the middle of the encoder model. The term middle means the most optimum place for the decoder to generate the colorized image which was found by hyperparameter tuning. Finally, the 2 paths are fused and sent to the decoder model.

D. Accuracy Analysis

Accuracy analysis is a crucial step in evaluating the effectiveness of colorization models. Different metrics help in comparing the predicted and original images. Mean Absolute Error (MAE) assesses the absolute difference between predicted and ground truth RGB values per pixel separately in red, green, and blue channels. Root Mean Square Error (RMSE), also known as average pixel error magnitude in

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RGB space, evaluates model performance by calculating the deviation between expected and actual pixel values, with lower values indicating better performance. MSE measures the squared variance between predicted and original image values averaged across all pixels in the image. Lower MSE values suggest better predictions. PSNR measures image quality in decibels, with higher values considered to be better.

Heat maps visualize data using color to depict the magnitude of a value. This paper suggests the use of heat maps in assessing image colorization accuracy by displaying errors between predicted and ground truth colors, highlighting the regions of the image where the predicted differs the most from the ground truth color. The maximum error that can occur is 255 for a pixel based on a single channel. Hence, the colors used for the heat map are divided into eight categories. The comparison between predicted and original images are extended to a 3-dimensional space using RGB values. Each pixel has corresponding Original (PO) and Predicted (PP) RGB values, forming a 3-D error represented by a dashed line as shown in Figure 1. RGB Error Magnitude per Pixel measures the distance between the RGB values of each pixel, calculated by taking the Euclidean distance between the RGB values of each pixel in the original and corresponding predicted image as of equation (1).

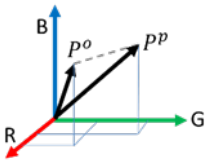


Figure 1. RGB values in 3D space for a pixel

$$\text{RGB Er. Mag. Per Pixel} = \frac{|R^o_{i,j} - R^p_{i,j}| + |G^o_{i,j} - G^p_{i,j}| + |B^o_{i,j} - B^p_{i,j}|}{3} \quad (1)$$

III. Results And Analysis

Although multiple parameters were analyzed for comparison, the paper selectively presents only a subset to ensure clarity and ease of comprehension for the reader.

A. Models used for Colorization

The results from MAE, PSNR values, and the average error magnitude per pixel of the predicted images of GAN and AE colorization models are compared for each Test 1 and Test 2. Comparison of MAE in R, G, and B channels for Test 1 and comparison of PSNR values for Test 1 between AE and GAN models are depicted in Figure 2. Green error consistently remains lowest, while blue error is consistently the highest. Doubling GAN epochs from 500 to 1000 significantly reduced blue error relative to R and G. Overall, AE model exhibits better results than GAN model, leading to its selection for further research analysis.

B. Effect of Training Batch Size and Iterations

For CNN AEs, Stochastic Gradient Descent (SGD) and Adam are common optimizers [32]. However, due to advantages over other optimization techniques, Adam is used in this application. The AE model was evaluated using different batch sizes, affecting the number of iterations.

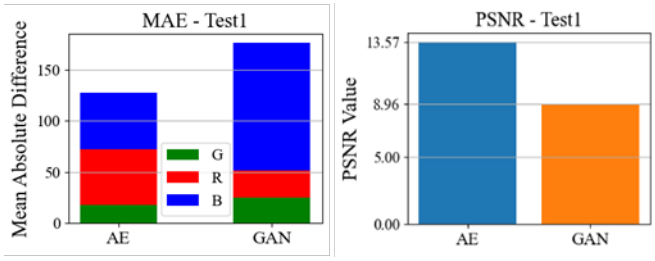


Figure 2. Comparison of GAN and AE models

Figure 3 shows the outcome of these tests: "Model 1" corresponds to training with 32 images per batch for one iteration, "2" involves 16 images per batch for two iterations, and "3" includes 340 images per batch for 20 iterations. The analysis of normalized MSE and PSNR values shows that configuration "1" performed better than "2," indicating that a batch size of one iteration may enhances accuracy:

- A larger batch size exposes the model to more diverse examples per iteration, aiding better generalization and improving its ability to handle unseen data.
- With larger batches, the model updates its weights based on gradients computed from a more representative sample. This reduces the impact of noise in smaller batches, resulting in a smoother estimation that helps the model converge more effectively toward the optimal solution, potentially improving accuracy.

C. Effect of Convolutional Layers, Filters, and Output Sizes

The convolutional layers, filters, and output sizes are fundamental components of CNNs. Figure 4 shows the results and the comparison of these three parameters by using the four models. Model 1, the basic AE model, served as the reference, produces an output size of 32x32. It has a maximum of 512 filters for a layer, 8 convolutional layers in the encoder and 5 in the decoder. The encoder layers are increased to 9 and the decoder to 7 in Model 2. The output size remained the same. Model 3 retained the layer configuration of Model 2 but reduced the output size of the encoder to 16x16. Finally, it was decided to change all the parameters, increasing encoder layers to 10, reducing decoder layers to 7, output size to 8x8, and with a maximum of 1024 filters in Model 4.

In Figure 4, normalized MSE Test 1 shows that models 2 and 3 have the lowest values. While model 2 records the absolute lowest, model 3 (0.0276), with a negligible difference of 0.0005, can also be considered as having the lowest value for MSE Test 1. In PSNR Test 1, model 3 records the highest value at 16.1.

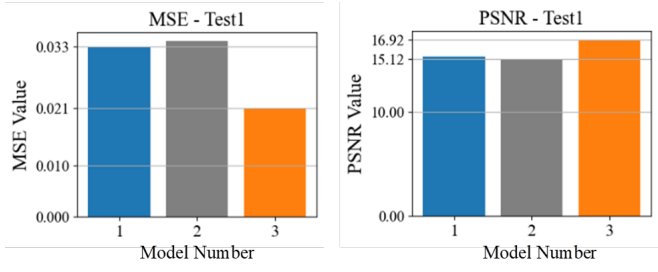


Figure 3. Training batch sizes and iteration results

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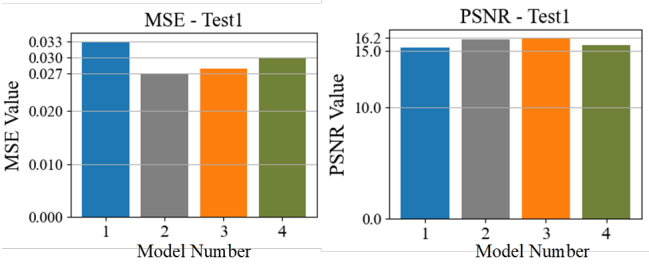


Figure 4. Convolutional layers, filters, and output sizes results

Thus, model 3 is identified as the best-performing model. With the results obtained, increasing the number of layers has a positive but limited impact on greyscale image colorization accuracy. Similarly, reducing the encoder output size has a positive, yet modest, effect on accuracy.

D. *Improved Algorithm*

Based on the comprehensive analysis previously described, the improved algorithm was developed. Figure 5 displays predicted images from the improved algorithm, while Figure 6 shows predictions from the basic convolutional AE model, with all other parameters held constant. A qualitative comparison reveals the following observations:

- Figure 5 exhibits higher feature extraction, evident in the absence of patches in the sky, and greenery areas compared to Figure 6.
- The overall color saturation is also higher in Figure 5, than Figure 6, making it look more vibrant.

To obtain a more generalized metric for comparison, both the improved algorithm and the basic algorithm were tested on 20 related images. Table I gives a comparison of the normalized MSE and PSNR values. The Improved Algorithm is better with lower MSE (0.0145) and higher PSNR (18.6291) compared to the Basic Algorithm. The average Error Magnitude per pixel is also calculated for each model. It is 30.54 for the Basic Convolutional Autoencoder and 25.09 for the Improved Algorithm. The smaller error magnitude per pixel in the Improved Algorithm indicates the enhanced accuracy in prediction. This improvement has been visually confirmed by demonstrating the correction of patches present in the basic model and advancing the goal of improved grayscale image colorization. Finally, Figure 7 shows the Improved Algorithm with separate blocks for convolutional layers, and Table II provides the model summary.

The algorithm combines several special features to enhance performance: LeakyReLU for continuous learning by allowing small gradients when inactive, while BatchNormalization stabilizes and accelerates training by normalizing inputs. Conv2DTranspose improves upscaling by learning filters during training, enhancing image reconstruction quality. Additionally, an effective fusion technique is used which combines deep and shallow features using Concatenate.

E. *Error Analysis*

However, relying solely on various metrics for image colorization, which aims to produce plausible colours rather than restore them, may have limitations since visual appeal and realism might not align with metric performance [11]. Therefore, this research introduces heat maps to analyse accuracy and characteristics of the predicted output for any AI model. Figure 8 shows the results of the improved algorithm, where the grayscale input is processed by the trained model, producing a predicted image. For accurate analysis, the original image is saved separately. RGB error magnitude is computed by comparing the predicted image with the original, analyzing individual R, G, and B channels. This generates the R, G, and B error magnitude per pixel. While qualitative visual representations of predicted images are common, quantitative visual illustrations of error analysis are lacking in the literature. Hence, introducing heat maps to show RGB, R, G, and B error magnitudes per pixel is a novel concept.

TABLE I. COMPARISON OF MSE AND PSNR VALUES

Model	MSE	PSNR
Improved Algorithm	0.0145	18.6291
Basic Algorithm	0.0242	16.4858



Figure 5. Predicted images from the improved algorithm

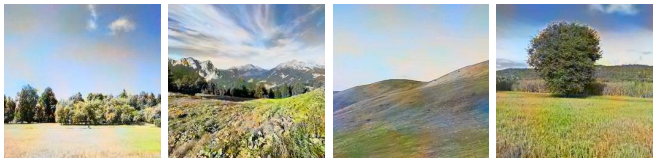


Figure 6. Predicted images from the improved algorithm

F. *Model Comparison*

This research compares the improved algorithm with the basic or vanilla convolutional autoencoder. MSE, PSNR, and average Error Magnitude per pixel have been used as comparison metrics. Both models were trained on the same dataset, facilitating a direct comparison. However, comparing this improved algorithm with other existing models poses challenges due to difficulties in replicating their exact model configurations and thereby being unable to train on the same dataset. Whereas, comparing the improved algorithm with the existing models is doubtful due to its limited training on only 346 images, which may not provide a fair representation of its performance against models trained on much larger datasets as mentioned in the introduction section.

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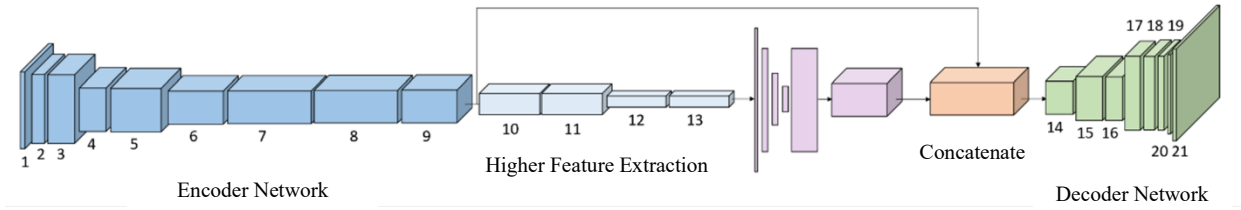


Figure 7. Improved Algorithm

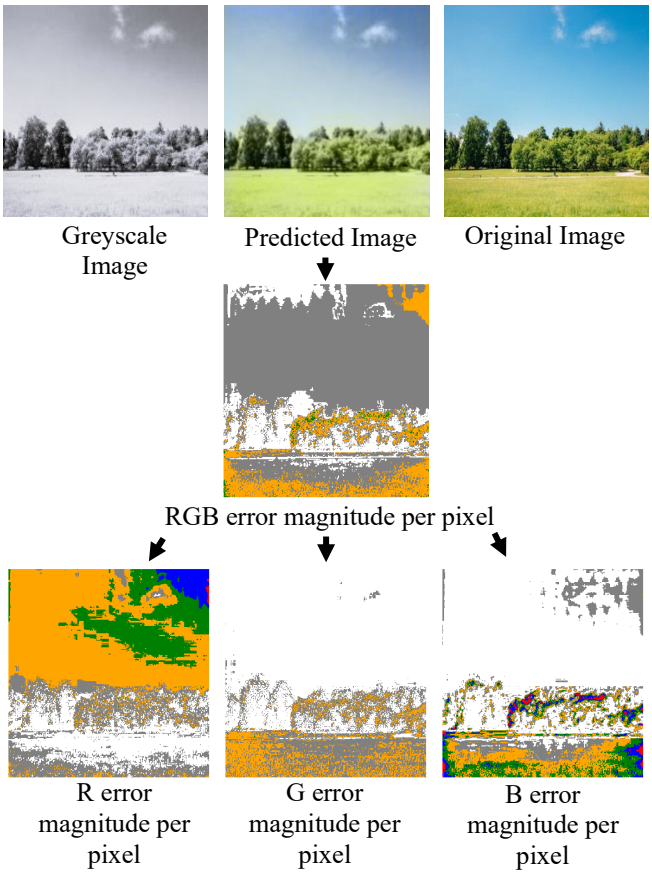


Figure 8. Heat maps for error identification

IV. Conclusion

In conclusion, this paper introduces an enhanced algorithm for automatic colorization of greyscale images through a deep-learning fusion technique. The Average Error Magnitude in a Pixel, MSE, and PSNR values for the Improved Algorithm compared with Basic Convolutional Autoencoder indicate better accuracy and colorization. Unlike training millions of images for weeks with pre-trained models, this is achieved by minimum computational requirements. Additionally, this level of accuracy and plausible colorized pictures were predicted from just 346 images for 200 epochs. Moreover, visual comparisons reveal the Improved Algorithm produces superior colorization, minimizing patches and staying within image feature boundaries. Moreover, a new quantitative technique using heat maps is proposed to visually illustrate errors in predicted images across various AI models.

TABLE II. Improved Algorithm Model Details

No.	Type	Stride	Output Size	Features
1	Input	-	256 x 256	1
2	Conv.	2 x 2	128 x 128	64
3	Conv.	1 x 1	128 x 128	128
4	Conv.	2 x 2	64 x 64	128
5	Conv.	1 x 1	64 x 64	256
6	Conv.	2 x 2	32 x 32	256
7	Conv.	1 x 1	32 x 32	512
8	Conv.	1 x 1	32 x 32	512
9	Conv.	1 x 1	32 x 32	256
10	Conv.	2 x 2	16 x 16	256
11	Conv.	1 x 1	16 x 16	256
12	Conv.	2 x 2	8 x 8	256
13	Conv.	1 x 1	8 x 8	256
14	Conv.	-	32 x 32	128
15	UpS.	-	64 x 64	128
16	Conv.	-	64 x 64	64
17	UpS.	-	128 x 128	64
18	Conv.	-	128 x 128	32
19	Conv.	-	128 x 128	16
20	Conv.	-	128 x 128	2
21	UpS.	-	256 x 256	2

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